

to T.575/05 and argued that in a similar situation a board had decided that although standards were available, the patent did not provide sufficient guidance as to which one should be used (invention insufficiently disclosed). However according to the board in T.1960/14, unlike in T.575/05, where many commonly used but not clearly defined methods of determining a parameter were available and thus led to a high degree of uncertainty for the skilled person trying to carry out the invention, at the priority date of the patent in suit there were only three clearly defined standard methods of determining the melting point of palm oil fractions, which produced very similar results with a fairly minor degree of ambiguity/uncertainty only at the margins of the claimed range. The issue of Art. 84 EPC was also addressed (T.256/87 and T.815/07 not followed).

In T.786/15 numerous methods of measuring the parameter Tg existed and the skilled person was aware of three ways, but the patent did not indicate any method for its measurement. To determine whether the Tg parameter was "so ill defined", it had to be investigated which methods would be considered by the skilled person. Referring to T.608/07 (point 2.5.2 of the Reasons), the board concluded that the alleged ambiguity arose at the edges of the values recited in claim, but this in itself could not lead to a finding of insufficiency of disclosure.

See also in this chapter: T.492/92 (two methods suggested by applicant) and T.1414/08 (parameter tensile strength – paper - not considered unusual – but no specific method of measuring disclosed – several standardised test methods in the art – consequences – end values ambiguous – scope – issue of Art. 84 EPC).

c) No or incomplete information on method of determining parameter

Where the calibration of (undisclosed) test conditions may be achieved although the methods of determination of the parameter are incompletely described, the invention may be sufficiently disclosed. See for example T.1062/98. On calibration, see also chapter II.C.6.6.8.

The fact that no direct independent method of specifically determining the parameter has been described is not in itself prejudicial to the sufficiency of the description where the claims do not relate to a method of determining the parameter (T.256/87, a case concerning Art. 84 and 83 EPC, with respect to an amended claim).

In T.83/01 (essential parameter –diameter of water droplets in a plastic fat-continuous spread – particularly difficult to measure) the board held that where the skilled person had no reason to doubt the definition of the parameter given, but there was no indication in the patent how to measure this parameter, the patent failed to fulfil the requirements of Art. 83 EPC 1973.

The board in T.2403/11 (see catchword) acknowledged that an ambiguity of a parameter in the claim was not enough in itself to deny sufficiency of disclosure. Whether such an ambiguity led to insufficiency of disclosure was to be decided on a case-by-case basis (see T.593/09 and T.472/14). The case at issue was different from T.882/03, which also concerned viscosity and where the variations resulting from the ambiguity were only minor,

and from T 492/92, where the skilled person knew which method to choose. In T 2403/11 the method and the measurement parameters to be chosen were not known to the skilled person. The board in T 466/05 (also relating to viscosity) likewise drew a contrast with T 492/92 on the basis that the skilled person in the case before it did not even know which parameter (i.e. which measure of the molecular weight) of the claimed polysaccharide should be determined. For other decisions relating to viscosity, see e.g. T 805/93, T 808/09 and T 482/09.

According to T 492/92 (no need to specify one of the several available analytical methods): where it is obvious that a skilled person would select a particular analytical measuring method, (none being disclosed in the patent), balancing its simplicity and convenience against the required accuracy, the requirements of Art. 83 EPC are met. In T 492/92 it was considered that the fact that two methods suggested by the appellant did not necessarily lead to identical results when measuring a specific parameter was not sufficient evidence that a skilled person could not determine this parameter of the claimed compositions with the required accuracy.

In T 2096/12 the skilled person could not know, from the disclosure in the patent, which measurement method should be employed to establish the claimed thickness parameter. The board agreed with the statements in T 593/09 and emphasised that the absence of a test method for a parameter which was a claimed feature did not lead by itself to the issue of insufficient disclosure. In a case where, for example, ranges for length or width of a clearly structured article were concerned, the parameters could be established unambiguously and without doubt. However, in each case it had to be evaluated on a case by case basis whether this was possible. When the extent of the protection conferred by the patent is not defined and cannot be reliably determined – such as in the case at issue where neither the claims nor the description provided a clue for how to interpret the parametrical feature of the claim, it can only be concluded that the requirement of Art. 100(b) EPC was not met.

In T 1583/17 the invention concerned the use of coated films. As regards determining the thickness of the coating, it was undisputed that neither claim 1 nor the description indicated a method to be used for this and it was also undisputed that different methods were available to the skilled person and that these methods might produce different results. The mere fact that a claim was unclear or its scope ambiguous did not automatically mean that the invention it defined was not sufficiently disclosed. The board recalled the case law and observed that, in the majority of cases, the absence of an indication of a method for measuring a parameter had been regarded as a problem under Art. 84 EPC only. The board stated that the absence of an indication of a method for measuring the thickness did not prevent the skilled person from practising the claimed invention and was therefore not objectionable under Art. 100(b) EPC. As the claimed invention was not restricted to thicknesses measured by a particular method, the skilled person was free to use any suitable method. The selection of a suitable method did not involve an undue burden since measuring the thickness of a coating or of a layer in general was an absolute standard procedure for which many commonly known methods were available. The board observed that the invention would possibly **not be sufficiently disclosed** if it could only be carried out with coatings having a thickness **measurable only with a specific, yet undisclosed**

method. The board stressed again that no specific degree of soft touch was required according to the claimed subject-matter (a crucial difference from T.225/93). The board concluded that there was no evidence on file that showed that the absolute thickness of the coating and thus the selection of a method for measuring the thickness was critical for carrying out the invention, i.e. for producing a coating having a thickness in the claimed range and a certain degree of soft touch.

In T.1064/15 the person skilled in the art was faced with the undefined parameter "diameter (SD)" and did not know how to choose the cross-section in order to obtain the desired technical effect. For the conditions of sufficiency of disclosure to be fulfilled it is not enough to be able to manufacture an object falling under the wording of a claim. That object must also exhibit the alleged or desired technical effect obtained with that invention (T.815/07). The field of non-circular cross-sections is far wider than that of circular ones, since it encompasses an enormous variety of shapes. This made it even more important to know how the key parameter of such shapes, namely the diameter, was determined.

In T.875/16 (optical control of hair growth), the board stated that the concept of duration of the pulse, as it appeared in claim 1, was vague and its use affected the claim's clarity. While the presence of this unclear parameter in the claim was not, in itself, sufficient to show that the skilled person could not reproduce the claimed device without undue burden, its presence was not without consequences for the issue of sufficiency, since the skilled person would not be able to rely on a clear definition of the parameter when attempting to reproduce the invention over the whole ambit of the claim. As to the discussion in relation to the whole ambit of the claim, the board found, in particular, that the effect – a degree of reduction in growth – was not specified and there was no indication as to how it was to be measured. This lack of any indication of how the claimed effect was measured was particularly problematic since it constituted the technical contribution of the invention over known devices.

In T.417/13 (PVC particles) the size of the PVC particles was an important feature of the invention. A skilled person needed to be capable of establishing which PVC particles had the appropriate size, as defined in the claims. The results for the particle size might vary to a large extent depending on the method of measurement. The description of the application provided only very limited information that could lead to the selection of a particular method of measurement. The person skilled in the art had thus to rely on his common general knowledge for the selection of an appropriate method of measurement. A person skilled in the art would be aware of the existence of the ISO standards. Thus, contrary to the situation in T.225/93 (no indication of suitable method), the skilled person in T.417/13 would have chosen a particular method (no evidence supplied by opponent on file as to the effect of measurement conditions on whether Art. 83 EPC requirements met).

In T.1900/17 the board decided after detailed reasoning that the ambiguity, due to the lack of definition of the method of calculation of the claimed parameter (calculated ClogP value for a nonionic surfactant), had not been shown to lead to an insufficiency, not even by following the rationale of T.1845/14.

T.2399/10 concerned a case where a starting material could not be provided, not only because no method was described, but also because it was characterised by an undefined parameter and therefore could not be identified. The patent thus did not disclose how to produce the alumina particles needed to obtain the claimed composition. This situation differed from cases where the features of the end product described in the patent were not clearly defined, which frequently resulted in a lack of clarity. Instead, it was a starting material which had not been here defined clearly enough. Since there was therefore inadequate information to select the **starting material**, the claimed composition could not be produced. This typically meant there was insufficient disclosure. (On disclosure of a starting material, see also **T.1596/16**).

See also **T.1305/05** and **T.1553/16** (unusual parameter – method of measurement kept secret) in this chapter II.C.5.3.

d) Effect of ambiguous parameter on claim interpretation

Where inventions are defined by claims containing unclear features, e.g. **ambiguous parameters**, which cannot be clarified in the light of the description, the effect sought may be the only means of understanding such inventions. This effect must then be taken into consideration and assessed under **Art. 83 EPC** with a view to establishing whether the invention can be reproduced (see **T.862/11**).

According to **T.1845/14** (two contradictory definitions of a parameter): The ambiguous definition of a parameter in a claim may result in the scope of the claim to be broader than the patentee might have intended. In such a case the question arises whether the teaching of the patent in suit, which was directed to the claimed subject-matter having regard to a specific meaning of that parameter (which, however, was omitted), would nevertheless have enabled the skilled person to carry out the invention outside of the scope intended by the patentee, using common general knowledge and a reasonable amount of experimentation.

In **T.1845/14** the board stated that the invention for which sufficiency of disclosure had to be assessed is defined by the terms of the claims, whose meaning had to be construed in view of the ambiguous definition of the parameter (CDBI). Following the normal rule of claim construction, terms used in a claim, such as parametric definitions, should be given their broadest technical sensible meaning in the context in which they appear and having regard to the common general knowledge. Concerning the CDBI, there was no evidence that it had a recognised meaning in the art at the priority or filing date of the patent in suit and the contradiction between the two definitions of it provided in the specification could not be resolved. Thus, this wording must be broadly construed to cover both definitions provided in the specification. Then the ambiguous definition of a parameter in a claim may result in the scope of the claim to be broader than the patentee might have intended. It was undisputed that the whole teaching provided in the specification related to only one definition of CDBI provided in the specification. Even accepting that the teaching was sufficient for one of the two definitions, the patent proprietor did not indicate how the skilled person, based on their common general knowledge, would have been in a position to complement the teaching contained in the patent in suit to prepare with reasonable effort

the copolymers in accordance with the second definition of the CDBI which did not belong to the group of polymers in accordance with the first definition of CDBI. The subject-matter of claim 1 has not been shown to be enabled in the whole range claimed, and the main request was refused (Art. 100(b) EPC).

In T 482/09 (viscosity) the boards held that, in opposition and any ensuing appeal proceedings, the result of using such an indefinite term in a claim was that it could not be taken to determine the limits to the protection sought, so that novelty and inventive step had to be assessed in the light of the prior art identified on the basis of all its technically meaningful possible interpretations. If the term had no specific meaning whatsoever, it would even lose entirely its effect of delimiting the claimed subject-matter from the relevant prior art.

5.5.2 Open-ended parameter features

In T 1697/12 the claims covered, through open-ended ranges, embodiments that could not be obtained with the process disclosed in the patent, but which might be obtainable with different methods still to be invented in the future (insufficiently disclosed invention). Related decisions: T 61/14 (point 5 of the Reasons, along the same lines, range not achievable over whole scope), T 2344/12 (point 1.1.2 of the Reasons, along the same lines, open ended range not sufficiently disclosed), T 517/98 (point 1.5 of the Reasons, similar, no teaching for whole range), T 615/19 (point 1.2 of the Reasons, conclusions of T 1697/12 not applicable, implicit limitation of open-ended range due to the existence of other features) and T 398/19, summarised below, in which the board rejected objections to **open-ended ranges** starting with "at least ..." or "at most ..." which the opponent had raised in relation to preparation methods that were still to be invented, invoking T 1697/12 and T 113/19 and the German Federal Court of Justice's decisions X ZR 32/17 and X ZR 34/17 in support of its arguments.

The decision in T 398/19 contains detailed reasons on the issue of a parameter defined by open-ended ranges. The opponent had objected to open-ended ranges of the kind starting with "at least ..." or "at most ...", alleging that they were not sufficiently disclosed. The board observed that the sole fact that certain features were expressed in terms of open-ended ranges was not enough to support an objection of insufficient disclosure, especially where, as in the case in hand, the description and the examples contained precise information on how to prepare one or more specific products having all the claimed features. Besides, it was common practice in the technical field concerned to characterise a product in terms of open-ended ranges and the claimed subject-matter often omitted parameters equally important for the claimed product's efficacy. Defining a product by all its structural parameters would clearly be impossible, and indeed often unnecessary in view of the invention's purpose. Lastly, the board observed that defining a parameter by an open-ended range by no means precluded the later invention of new methods whereby parameters having values very far from those claimed and so likewise patentable could be obtained. The board found that the requirements of Art. 83 EPC were met, endorsing the findings in T 113/19, T 1018/05 and T 624/08, and holding T 1697/12 to be inapplicable to the case in hand.